

Estela Pérez Luque, PhD

Research Engineer

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PROFILE

I develop and evaluate computational models of behaviour in dynamic environments. I am especially motivated by understanding human behaviour and using that knowledge to improve how technology adapts and supports real-world interaction. My background combines predictive modelling, simulation, motion capture, and time-series analysis to translate theory into testable code and structured evidence. Most recently, I developed a novel 3D motion prediction pipeline by extending steering dynamics into 3D and integrating it with adaptive velocity control and inverse kinematics, then benchmarked it against experimental data to analyse both prediction accuracy and human variability. I am motivated by research problems where modelling, experimentation, and rigorous evaluation come together to make complex systems more adaptive, robust, and useful in real-world settings.

TECHNICAL SKILLS

Programming: MATLAB, Python, C++ (basic)

Modelling: Dynamic systems, inverse kinematics, optimization, simulation, time-series analysis

Research & validation: Comparative benchmarking, Experiment design, motion capture, model evaluation

Domain: Human behaviour modelling, motion prediction, digital human modelling

Additional tools: IPS IMMA, Siemens Jack, Autodesk Inventor, SolidWorks

Languages: English (C1), Spanish (Native), Swedish (B1-improving)

PROFESSIONAL EXPERIENCE

PhD Researcher - University of Skövde, Sweden (Onsite)

Sep 2020 – Oct 2025

- Built end-to-end MATLAB and Python pipelines to benchmark predictive models against motion capture and in-vehicle datasets, analysing datasets of up to 199 participants across multiple tasks and operational conditions, and generating reproducible, traceable evidence to support model validation and design decisions.
- Extended a steering dynamics model from 2D to 3D using quaternion algebra and integrated it with an adaptive velocity model and FABRIK inverse kinematics to develop a first-of-its-kind heuristic motion prediction pipeline for full 7-DOF upper-extremity trajectories.
- Designed and executed a comparative analysis of the heuristic method against an optimization-based baseline using motion capture data from 10 participants across 4 reach tasks, evaluating performance across path, velocity, and upper-extremity joint configuration.
- Introduced DTW alongside RMSE and developed an inter-subject variability analysis to assess trajectory shape, temporal alignment, and how closely prediction models reproduced the natural diversity present in observed human motion, identifying up to ~15–20% variation in model performance depending on evaluation approach.
- Designed and executed scenario-based validation studies comparing simulated and real-world behaviour across multiple system configurations and user groups, enabling structured model comparison and reducing reliance on subjective assessment in early design stages.
- Applied multi-objective optimization (NSGA-II) to evaluate 1,000+ vehicle interior configurations across 14 digital human models, transforming ergonomics assessment from manual trial-and-error into a scalable, data-driven design exploration process.
- Collaborated with Volvo Cars, Volvo Trucks, Scania, and the SAFER vehicle safety ecosystem, aligning validation strategies with engineering constraints and real-world use cases and enabling faster, more consistent decision-making based on data rather than assumptions through 20+ scientific publications and 15+ presentations (international conferences, industry workshops, and invited talks).

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Visiting Researcher - University of Antwerp, Belgium (Onsite)

Sep 2024 – Mar 2025

- Led full-cycle experimental data collection for a large-scale, multi-year study, including ethics approval, protocol design, participant recruitment, and multi-sensor data acquisition, enabling long-term, structured validation of human behaviour models across datasets and studies.
- Designed and executed motion capture and 4D body scanning protocols for 30+ participants, ensuring consistent data quality and reproducibility across experimental conditions, improving reliability of validation datasets used for model evaluation.
- Built structured, reproducible data pipelines to standardize multi-site data collection and processing; increased data consistency and comparability across institutions, supporting more robust cross-study validation.

Co-founder and R&D Lead - Swe-Ex AB, Sweden (Startup)

Sep 2021 – Mar 2023

- Co-founded a startup developing wearable safety support equipment for industrial environments, focusing on ergonomic risk reduction and worker support.
- Translated user needs and safety requirements into measurable design and validation criteria, guiding three full prototype iterations from CAD concept to physical validation using 3D-printed, textile, and steel components.
- Developed and tested prototypes using structured physical validation methods, enabling rapid iteration cycles and evidence-based design decisions.
- Built business and product understanding through a 2-year incubator program, gaining experience in prioritization under uncertainty, stakeholder communication, and fast-moving development environments.

Research Assistant - University of Skövde, Sweden (Onsite)

Jun 2019 – Sep 2020

- Developed and executed simulation studies and ergonomic analyses across multiple projects with 6+ industry partners, producing structured reports and live demonstrations, and enabling stakeholders to understand and act on simulation results in early design phases.
- Created educational materials and training content, including guided video tutorials, for industrial partners and students, supporting adoption of digital human modelling tools and improving usability and onboarding time, and increasing effective use of simulation tools in both academic and industry settings.

EDUCATION

PhD in Informatics, University of Skövde (Sweden)

Sep 2020 – Oct 2025

[PhD Thesis](#): Human Posture and Motion Prediction for Automotive Ergonomics Design

Focus: Computational modelling, simulation, human motion prediction, experimental validation, time-series

Master of Science in Intelligent Automation, University of Skövde (Sweden)

Aug 2018 – Jun 2019

Bachelor of Science in Mechanical Engineering, University of Skövde (Sweden)

Aug 2017 – Jun 2018

Degree in Mechanical Engineering, University of Málaga (Spain)

Aug 2012 – Jun 2018

RESEARCH PUBLICATIONS

Pérez Luque, E., Lee, S., Högberg, D., Yang, J., & Lamb, M. (2026). Predicting Human Upper Extremity Reaching Motions: Comparison of Optimization-Based Method and Heuristic Method. *International Journal of Human-Computer Interaction*. DOI: 10.1080/10447318.2026.2632154

Full publication list available at: [Google Scholar – Estela Pérez Luque](#)